

## Progesterone

### Test Details

#### METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

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#### RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

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### Use

Establish the presence of a functioning corpus luteum or luteal cell function; confirm basal body temperature measurements for the occurrence of ovulation; obtain an indication of the day of ovulation; evaluate the functional state of the corpus luteum in infertility patients; assess placental function during pregnancy; ovarian function test

### Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

## Limitations

As with all tests containing monoclonal mouse antibodies, erroneous findings may be obtained from samples taken from patients who have been treated with monoclonal mouse antibodies or who have received them for diagnostic purposes.<sup>1</sup> In rare cases, interference due to extremely high titers of antibodies to streptavidin and ruthenium can occur.<sup>1</sup> The test contains additives that minimize these effects.

## Custom Additional Information

Progesterone is a steroid hormone with a molecular weight of 314.5 daltons.<sup>2</sup> Progesterone is mainly formed in the cells of the corpus luteum and during pregnancy in the placenta. Progesterone is increased in congenital adrenal hyperplasia due to 21-hydroxylase, 17-hydroxylase, and 11- $\beta$ -hydroxylase deficiency. Progesterone is decreased in primary or secondary hypogonadism and short luteal phase syndrome.

The progesterone concentration correlates with the development and regression of the corpus luteum. Whereas progesterone is barely detectable in the follicular phase of the female cycle, a rise in the progesterone level is observed one day prior to ovulation. Increased progesterone synthesis occurs during the luteal phase. In the second half of the cycle pregnanediol is excreted in urine as the main degradation product of progesterone.

Progesterone brings about the conversion of the uterine mucosa into a tissue rich in glands (secretion phase), in order to prepare for the intrauterine implantation of the fertilized ovum. During pregnancy, progesterone inhibits the contraction of the myometrium. In the mammary gland, progesterone (together with estrogens) promotes the proliferation and secretion disposition of the alveoli.<sup>2,3</sup>

The determination of progesterone is utilized in fertility diagnosis for the detection of ovulation and assessment of the luteal phase.<sup>3,4</sup>

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# Specimen Requirements

## Specimen

Serum (preferred) **or** plasma

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## Volume

1 mL

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## Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

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## Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

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## Collection Instructions

If a red-top tube or plasma is used, transfer separated serum or plasma to a plastic transport tube.

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## Stability Requirements

| Temperature        | Period    |
|--------------------|-----------|
| Room temperature   | 15 days   |
| Refrigerated       | 15 days   |
| Frozen             | 15 days   |
| Freeze/thaw cycles | Stable x3 |

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## Reference Range

See table.

| Gender/Stage     | Range (ng/mL) |
|------------------|---------------|
| <b>Male</b>      | 0.0–0.5       |
| <b>Female</b>    |               |
| follicular phase | 0.1–0.9       |
| luteal phase     | 1.8–23.9      |
| ovulatory phase  | 0.1–12.0      |
| <b>Pregnant</b>  |               |
| 1st trimester    | 11.0–44.3     |
| 2nd trimester    | 25.4–83.3     |
| 3rd trimester    | 58.7–214.0    |
| postmenopausal   | 0.0–0.1       |

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## Storage Instructions

Room temperature

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## References

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Stewart MO<sup>1</sup>, Whittaker PG, Persson B, Hanson U, Lind T. A longitudinal study of circulating progesterone, oestradiol, hCG and hPL during pregnancy in type 1 diabetic mothers. *Br J Obstet Gynaecol.* 1989 Apr; 96(4):415-423. PubMed 2751954 (<http://www.ncbi.nlm.nih.gov/pubmed/2751954>)

## Footnotes

1. Progesterone on Elecsys 1010/2010 and Modular Analytics E170, 2007-09, V 11 [package insert] Indianapolis, Ind: Roche Diagnostics; 2007.

2. Johnson MR, Carter G, Grint C, Lightman SL. Relationship between ovarian steroids, gonadotrophins and relaxin during the menstrual cycle. *Acta Endocrinol.* 1993 Aug; 129(2):121-125. PubMed 8372595 (<http://www.ncbi.nlm.nih.gov/pubmed/8372595>)

3. Runnebaum B, Rabe T. *Gynäkologische Endokrinologie und Fortpflanzungsmedizin*. Berlin, Germany: Springer Verlag 1994, Band 1: 36-38, 70, 116; Band 2: 137, 360, 398-399, 408-409, 422-423.

4. Guillaume J, Benjamin F, Sicuranza B, Wang CF, Garcia A, Friberg J. Maternal serum levels of estradiol, progesterone and human chorionic gonadotropin in ectopic pregnancy and their correlation with endometrial histologic findings. *Surg Gynecol Obstet.* 1987 Jul; 165(1):9-12. PubMed 3589936 (<http://www.ncbi.nlm.nih.gov/pubmed/3589936>)